AI-Powered Healthcare Q&A Chatbot

A Mini Project report submitted to

NARAYANA ENGINEERING COLLEGE (AUTONOMOUS), NELLORE

(Affiliated to Jawaharlal Nehru Technological University, Anantapur, Ananthapuramu)

In partial fulfilment of the requirements for the award of Degree of

Bachelor of Computer Science and Engineering Cluster Branches (CSM)

BY

VALLEPU VAMSI	(23711A3363)
GUNDLA SAI SRINATH	(23711A3321)
PESALA NIKHIL	(23711A3348)
PYDIMARRI SAI SAKETH	(23711A3351)
VEMURI OMESH ROHAN	(23711A3366)

Under the guidance of

Mr. V. SUDHAKAR M.Tech , (Ph.D)

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING CLUSTER BRANCHES

December, 2025

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CLUSTER BRANCHES

CERTIFICATE

This is to certify that the mini project report entitled **"AI-Powered Healthcare Q&A Chatbot "** submitted by **VALLEPU VAMSI (23711A3363), GUNDLA SAI SRINATH (23711A3321), PESALA NIKHIL (23711A3348), PYDIMARRI SAI SAKETH (23711A3351) and VEMURI OMESH ROHAN (23711A3366),** of the Department of Computer Science and Engineering (Cluster Branches) CSM, Narayana Engineering College (Autonomous), Nellore, is a bona fide record of the work carried out by them under my guidance and supervision during the academic year **2025-2026**

Signature of the Guide
Mr.V. SUDHAKAR M.Tech, (Ph.D)
Assistant Professor
Department of CSE
Narayana Engineering College
Nellore- 524004

Signature of the HOD
Dr. C Rajendra M.Tech., Ph.D.
Professor & HOD
Department of CSE Cluster
Narayana Engineering College
Nellore- 524004

NARAYANA ENGINEERING COLLEGE (AUTONOMOUS) NELLORE

VISION OF INSTITUTION

To be one of the nation's premier Institutions for Technical and Management Education and a key contributor for Technological and Socio-economic Development of the Nation.

MISSION OF INSTITUTION

Requirements & actions are identified to achieve the vision

M1	To produce technically competent Engineers and Managers by maintaining high academic standards, world class infrastructure and core instructions.
M2	To enhance innovative skills and multidisciplinary approach of students through well experienced faculty and industry interactions.
M3	To inculcate global perspective and attitude of students to face real world challenges by developing leadership qualities, lifelong learning abilities and ethical values.

VISION OF DEPARTMENT

To be a choice for education in the area of Computer Science and Engineering, serve as a valuable resource for IT industry & society and exhibit creativity, innovation and ethicsto cater the global challenges.

MISSION OF DEPARTMENT

Requirements & actions are identified to achieve the vision

M1	To educate learners by adapting innovative pedagogies for enhancing their cognitive skills, technical competence and lifelong learning.
M2	To provide training programs and guidance to learners through industry institute partnerships, social awareness programs, internships, competitions and project works to inculcate research skills to address the global challenges.
M3	To provide opportunities for students to practice professional, social and ethical responsibilities using IT expertise with a blend of leadership and entrepreneurial skills.

Department of

COMPUTER SCIENCE & ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO-1 Procure employment/progress towards higher degree and practice successfully in the CS/IT profession.

PEO-2 Address complex problems by adapting to rapidly changing IT technologies.

PEO-3 Gain respect and trust of others as effective and ethical team member by demonstrating professionalism and functioning effectively in team-oriented and open-ended activities in industry and society.

PROGRAM OUTCOMES (POs)

PO 1 : Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2 : Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3 : Design/Development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4 : Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5 : Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6 : The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7 : Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8 : Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9 : Individual and Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO-1 Domain Specific Knowledge: Apply the relevant techniques to develop solutions in the domains of algorithms, system software, computer programming, multimedia, web, data and networking.

PSO-2 Software Product Development: Apply the design and deployment principles to deliver a quality software product for the success of business of varying complexity.

NARAYANA ENGINEERING COLLEGE

ACKNOWLEDGEMENT

We thank our mini project supervisor **Mr. V.SUDHAKAR , M.Tech., (Ph.D). Assistant Professor**, Department of Computer Science and Engineering (CSE) Cluster, Narayana Engineering College, Nellore for his guidance, valuable suggestions and support in the completion of the project.

We would like to express our deep sense of gratitude and sincere thanks to **Dr. C. RAMA MOHAN, M.Tech., Ph.D**, Mini Project Coordinator, Department of Computer Science and Engineering (CSE) Cluster, Narayana Engineering College, Nellore for providing the necessary facilities and encouragement towards the mini project work.

We would like to express my deep sense of gratitude and my sincere thanks to **Dr. C Rajendra, M.Tech., Ph.D**, HOD, Department of **Department of Computer Science and Engineering (CSE) Cluster**, Narayana Engineering College, Nellore for providing the required and continuous encouragement during the successful completion of the mini project work.

We would like to express our profound gratitude to Principal **Dr. V. RAVIPRASAD, M.Tech., Ph.D**, Narayana Engineering College, Nellore for providing the required facilities.

We are thankful to our director **Dr. B. DATTATRAYA SARMA,** Narayana Engineering and Pharmacy Colleges who took keen interest and encouraged me in every effort throughout this course.

We are extremely grateful to **Sri. V. VIJAYA BHASKAR REDDY,** General Manager, Narayana Educational Institutions, Andhra Pradesh for their blessings.

We are extremely grateful to **Mr. K. PUNEET**, President, Narayana Educational Institutions, India for their coordination.

We are extremely grateful to **Dr. P. NARAYANA**, **Ph.D**, Founder, Narayana Educational Institutions, India for providing everything needed and supporting in everywhere.

We gratefully acknowledge and express our thanks to the teaching and the non-teaching staff of CSE department.

We would like to express our love and affection to our parents for their support and encouragement throughout this project.

With regards,

VALLEPU VAMSI (23711A3363)
GUNDLA SAI SRINATH (23711A3321)
PESALA NIKHIL (23711A3348)
PYDIMARRI SAI SAKETH (23711A3351)
VEMURI OMESH ROHAN (23711A3366)

DECLARATION

We hereby declare that the project entitled **AI-Powered Healthcare Q&A Chatbot** has been done by us under the guidance of **Mr. V.SUDHAKAR M.Tech, (Ph.D) , Assistant Professor**, **Department of Computer Science and Engineering (CSE) Cluster**. This Project work has been submitted to NARAYANA ENGINEERING COLLEGE, NELLORE as a part of partial fulfilment of the requirements for the award of degree of **Computer Science and Engineering Cluster Branches**.

We also declare that this project report has not been submitted at any time to another institute or University for the award of any degree.

Mini Project Associates

VALLEPU VAMSI (23711A3363)
GUNDLA SAI SRINATH (23711A3321)
PESALA NIKHIL (23711A3348)
PYDIMARRI SAI SAKETH (23711A3351)
VEMURI OMESH ROHAN (23711A3366)

Place: NELLORE

Date: